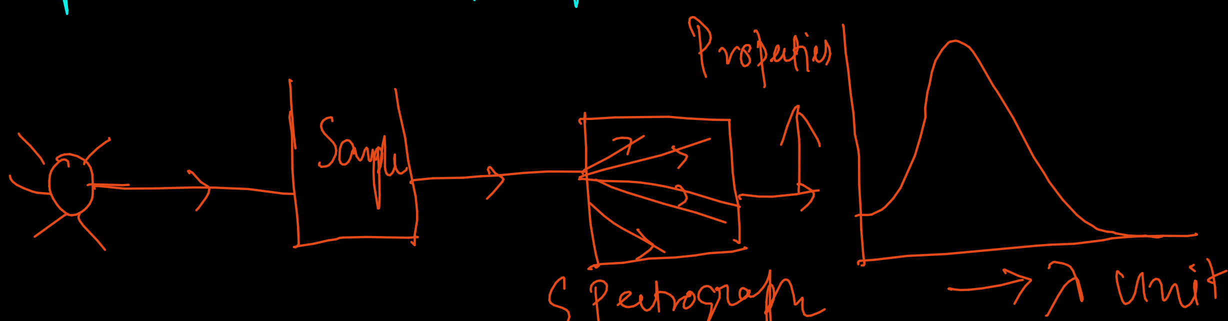
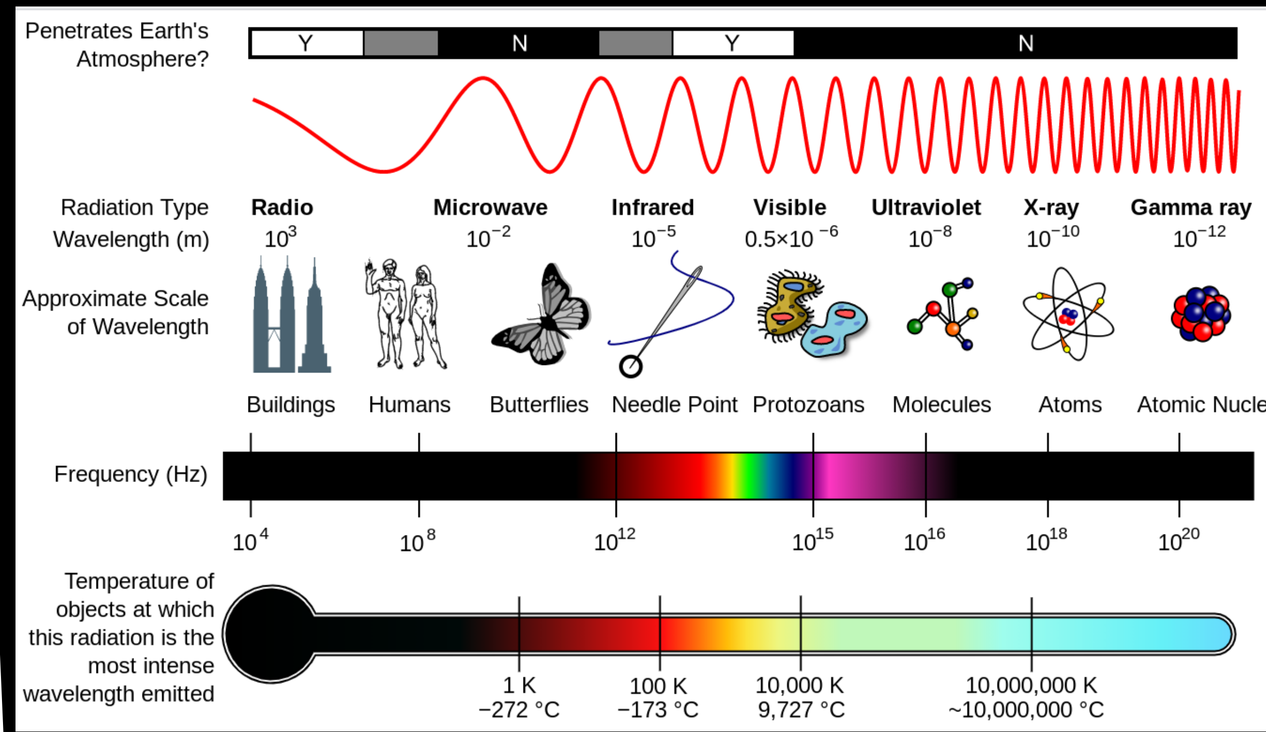
Spectroscopy

What is spectroscopy?

Study of interaction between matter and electromagnetic radiation

- Fundamental tool in the feild of physics, Chemistry, Astronomy, space science, moleculer engineering, nano-science
- Applied to study composite, physical structure, at atomic, molecular and micro/nano/macro scales. Molecular reaction, structure, dynamics, function can be done.
- Imaging of molecule, cell, tissue, medical Imaging etc.
- The device is known as Spectrograph, spectrometer, spectrophotometer, spectral analyzer

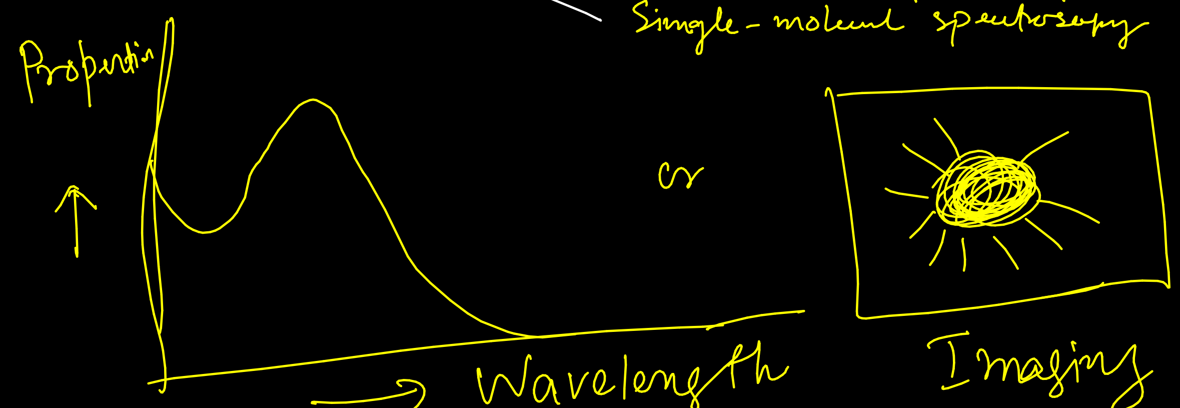




There are ~100s different types of spectroscopy: –

Mainly ⇒

1. Absorption spectroscopy → UV, UV+VIS, IR, Circular Dichroism | X-Ray
2. Emission Spectroscopy → Fluorescence, Raman Spectroscopy
3. Scattering Spectroscopy → Dynamic Light Scattering, Force Spectroscopy, Correlation Spectroscopy
4. Laser Spectroscopy → Time Resolved Spectroscopy, Single-molecule spectroscopy



Infrared (IR) Spectroscopy

⇓ condition

Molecule need to vibrate (ex. H – Cl)

⇓

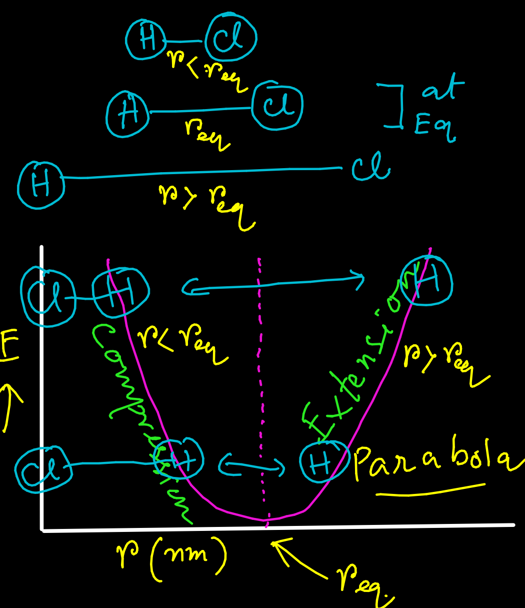
Should $\boxed{\text{change of dipole moment}}$ at the time of vibration

Restoring force, $f = -k(r - r_{eq})$

Because restoring force acts against the

Bond is like spring ⇒ obeys Hooke's Law

Energy of di-atomic molecule:



$$\boxed{\text{Energy, } E = \frac{1}{2}k(r - r_{eq})^2} \Rightarrow \text{Simple Harmonic Oscilator model}$$

Vibrational energy for a molecule is quantized ⇒ SHO model

$$E_v = \left(v + \frac{1}{2}\right)h\omega \text{ Joule}$$

v = vibrational quantum no. = 1, 2, 3.

$\bar{\omega}$ = oscillation frequency

$= \frac{1}{2\pi c}\sqrt{\frac{k}{\mu}}\ cm^{-1}$ $\left[\mu = \frac{m_1 m_2}{m_1 + m_2}\right]$

Oscillation frequency: $\omega \Rightarrow s^{-1}$

Wavenumber $\bar{\omega} \Rightarrow cm^{-1}$

Converting to spectroscopic unit:

$$E_v = \frac{E}{hc} = \left(v + \frac{1}{2}\right)\bar{\omega}\ cm^{-1}$$

At $v = 0$,

$$E_0 = \left(0 + \frac{1}{2}\right)\bar{\omega}\ cm^{-1} = \frac{1}{2}\bar{\omega}\ cm^{-1}$$

*** Zero-point energy

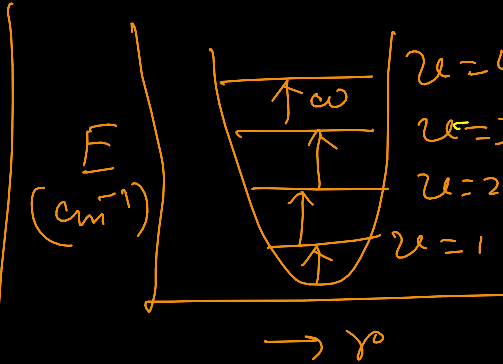
At $v = 1$, $E_1 = \left(1 + \frac{1}{2}\right) = \frac{3}{2}\bar{\omega}\ cm^{-1}$

At $v = 2$, $E_2 = \left(2 + \frac{1}{2}\right) = \frac{5}{2}\bar{\omega}\ cm^{-1}$

$\Delta E = E_1 - E_0 = \bar{\omega}\ cm^{-1}$

or $E_2 - E_1 = \bar{\omega}\ cm^{-1}$

• Vibrational states are equally spaced.



$$\boxed{\Delta v = \pm 1}$$

Selection Rule

Ref. Book: Fundamentals of Molecular Spectroscopy

Colin N. Banwell

Elaine M. McCash

IR Spectra of polyatomic molecule

CO_2, H_2O, SO_2

Degree of Freedom (DOF)

Total DOF $= 3N =$ Tra. + Rot. + Vib. (N = Total no. of atoms)

For H_2O, $N = 3$, DOF $= 3 \times 3 = 9$

	Linear	non-linear
• Translation	3	3
• Rotational	2	3
• Vibrational	$3N-5$	$3N-6$
└ Stretching	$N-1$	$N-1$ (no. of bond)
└ Bending	$2N-4$	$2N-5$

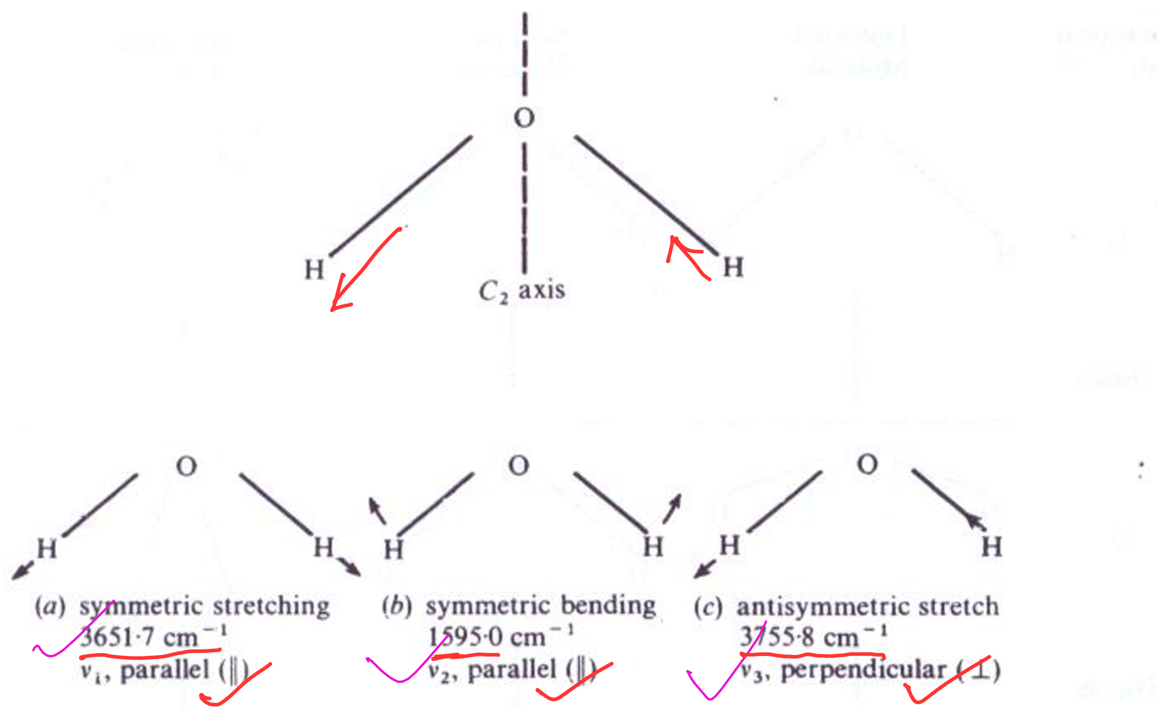


Figure 3.9 The symmetry of the water molecule and its three fundamental vibrations. The motion of the oxygen atom, which must occur to keep the centre of gravity of the molecule stationary, is here ignored.

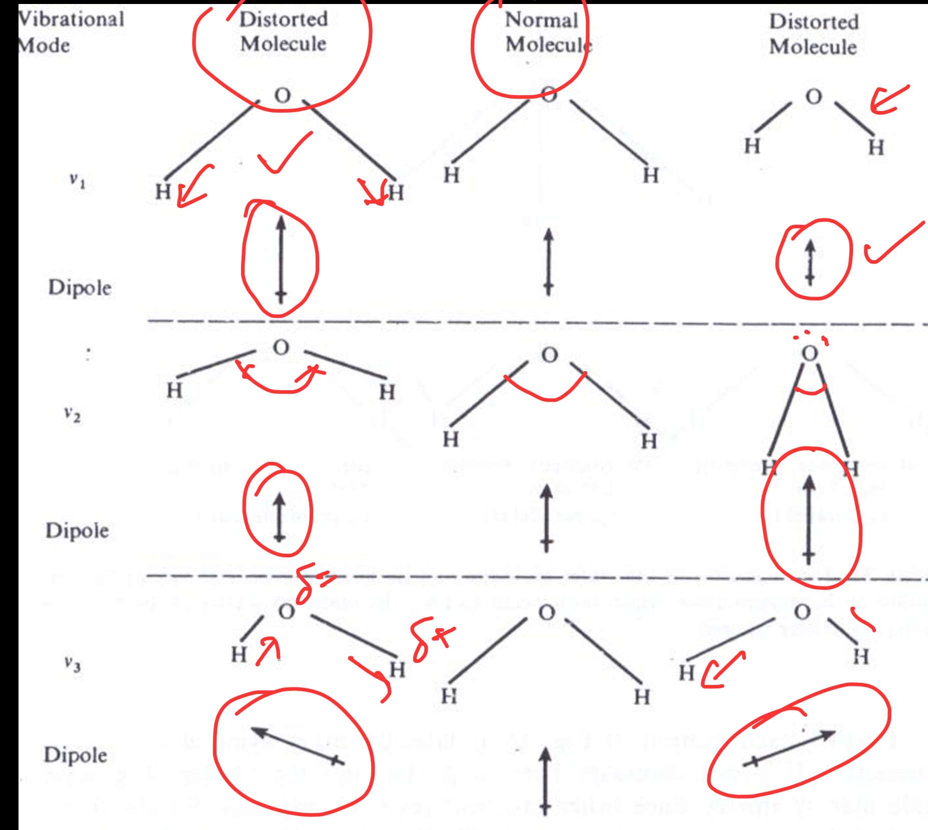


Figure 3.10 The change in the electric dipole moment produced by each vibration of the water molecule. This is seen to occur either along (∥) or across (⊥), the symmetry axis. The amplitudes are greatly exaggerated for clarity.

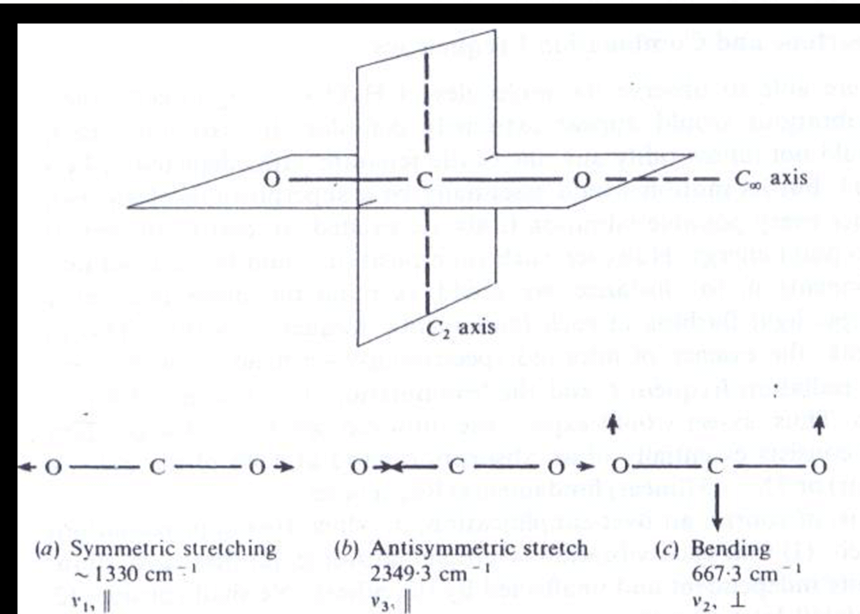


Figure 3.11 The symmetry and fundamental vibrations of the carbon dioxide molecule.

Data Analysis by IR Spectroscopy

① Skeletal vibration : $-C-O-C$, $-C-\overset{C}{\underset{C}{C}}-C$, 

Many space probe has IR spectrometer to identify compound on planet

1400 – 700 cm^{-1}

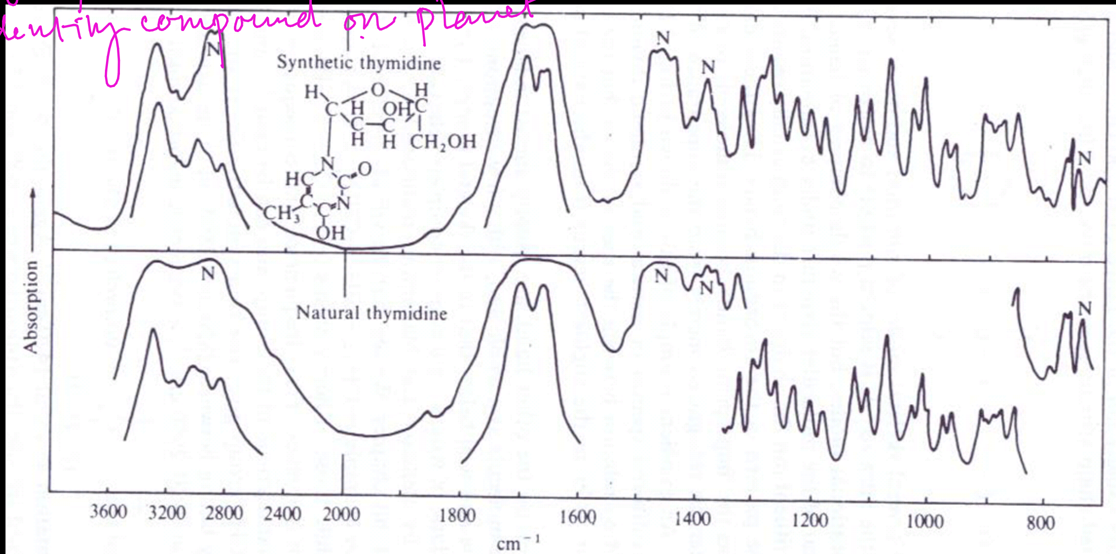


Figure 3.19(*a*) Comparison of the infra-red spectra of natural and synthetic thymidine, to show the use of the fingerprint region. (N = absorption from liquid paraffin (nujol) in which the solid thymidine is suspended.) *(These spectra and that of Fig. 3.19(b) are reproduced by kind permission of Professor N. Sheppard of the University of East Anglia, Norwich.)*

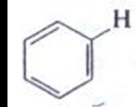
② Group frequency : $-CH_3$, $-OH$, $-C\equiv N$

$>C=O$, $-C-Cl$, $-C-Br$

$-\overset{O}{\overset{\|}{C}}-OH$, $-\overset{O}{\overset{\|}{C}}-O-Me$

$-Et$

Table 3.4 Characteristic stretching frequencies of some molecular groups

Group	Approximate frequency (cm^{-1})	Group	Approximate frequency (cm^{-1})
$-OH$	3600	$>C=O$	1750–1600
$-NH_2$	3400	$>C=C<$	1650
$\equiv CH$	3300	$>C=N-$	1600
	3060	$>C-C<$, $>C-N<$, $>C-O-$	1200–1000
$=CH_2$	3030		
		$>C=S$	1100
$-CH_3$	2970 (asym. stretch)		
	2870 (sym. stretch)	$>C-F$	1050
	1460 (asym. deform.)		
	1375 (sym. deform.)	$>C-Cl$	725
$-CH_2-$	2930 (asym. stretch)	$>C-Br$	650
	2860 (sym. stretch)		
	1470 (deformation)	$>C-I$	550
$-SH$	2580		
$-C\equiv N$	2250		
$-C\equiv C-$	2220		

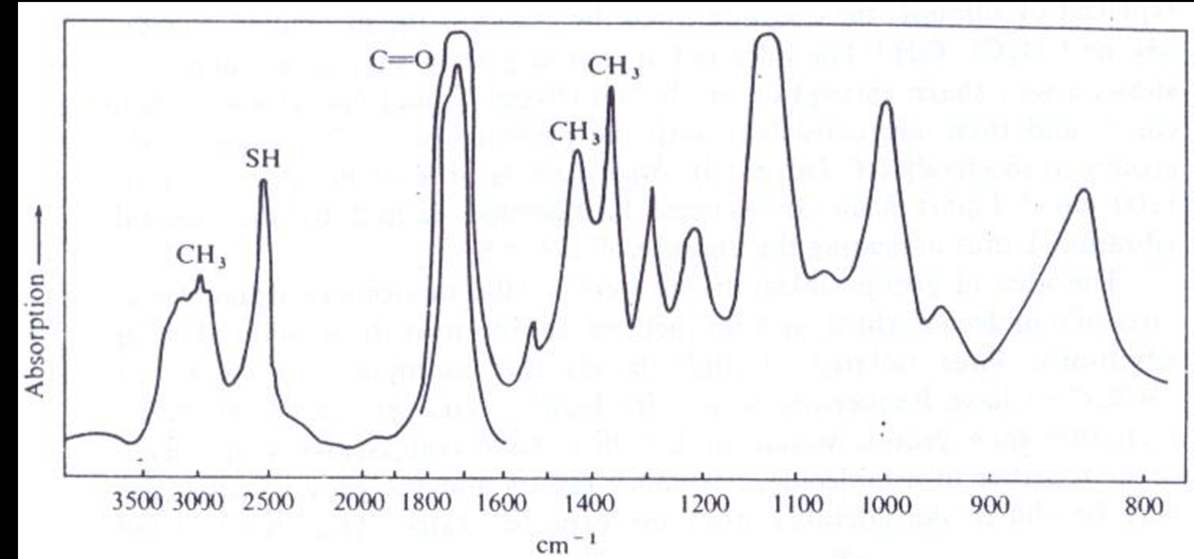


Figure 3.19(*b*) The spectrum of thiocetic acid, $CH_3CO.SH$.